Using UVEK with Premise

Progress review | 10 November 2026

Work is paused. The full project is not finished.

The aim

Let Premise use UVEK 2026 to describe how production might change in the future, while keeping the steps that users already follow. The work uses UVEK production datasets and the ecoinvent 3.10 list of emissions and natural resources.

What works now

Premise can update **biomass, electricity and heat** for the future scenario we tested: REMIND SSP2-NPi for 2050. It can then create a Brightway database containing **12,622 process datasets**. The links between datasets and the records of changes have been checked.

Part of the work	Current position
Electricity matching	All 42 difficult matching cases have been reviewed: 39 are handled, 2 are handled in part, and 1 is deliberately left out because no defensible replacement was found.
Biomass, electricity and heat	These three updates have been tested together, including creating and checking the Brightway database.
Other updates	The wider plan covers 19 updates. Only 3 are ready for the tested scenario. Emissions and 15 other updates still need work.
Heat coverage	Even within heat, 11 types of supply still lack a reviewed match. Other scenarios may need these, so the successful test does not establish that all heat cases work.

What has been preserved

The original dataset versions remain traceable. Checks cover input amounts, positive and negative signs, information about uncertainty, and links to the supplying processes. A matched case can still use an approximation; it does not mean the two databases describe exactly the same process.

Conversions between the cut-off and consequential approaches remain outside the agreed scope. These approaches make different choices about how production burdens and benefits are assigned.

Emissions: promising, but not ready for normal use

The reviewed emissions work covers **3,487 input entries**. It adjusts selected amounts of pollution released to air while keeping physical input quantities intact. The latest verified version passed 61 focused software tests and 196 calculations using positive and negative demands.

Those checks support the software implementation and its calculations. They do **not** establish that environmental impact scores are accurate, complete or equivalent to the original database. The emissions update is still unavailable through the normal user workflow.

Recent matching work changed configuration files rather than Python code. Earlier work also required changes to the program itself. The work is saved locally and has not been pushed.

What we should review next

Separate what works from the assumptions and unfinished work.

Scientific choices that matter

Negative treatment amounts. A minus sign on a waste-treatment input does not automatically mean an environmental benefit. The two databases may describe the same service using different sign conventions. We must check the meaning and reference quantity before interpreting the result; simply flipping the sign or removing the minus sign is not a reliable fix.

Approximations and missing inputs. Some matches use a process from another country, another technology or another production year. Two paperboard cases still leave out some parts. One electrical-steel input has no justified replacement. A separate solar-power case leaves out a claimed benefit from reusing nitrate salts. These choices are recorded, but their effects on impact scores have not been measured.

Pollution controls. The gas-supply work uses industrial gas combustion as a stand-in for some equipment, such as pipeline compressors. Pollution controls may develop differently in those uses. Two entries have an unresolved location, so a global trend is used without changing their recorded locations. The emissions step itself leaves carbon dioxide, particles, water and soil releases, and several other substances unchanged.

Why the latest batch is unfinished

The paused draft proposes **137 additional input entries across 25 process inventories**. Its small software tests pass, but a wider calculation check did not meet the required numerical accuracy. **The cause has not yet been identified.** These changes have not been accepted into the verified version and are excluded from the notebook's default setup.

Two further inputs also remain unchanged. Their source data combines methane from burning fuel with methane leaking from gas distribution. Applying one combustion adjustment to both would not be justified.

Three decisions for our review

1. Choose the next deliverable. Should we first consolidate the three working updates into a usable version, or give priority to adding more sectors? The broader goal remains paused, not cancelled.

2. Agree the acceptable approximations. Which substitute processes and missing inputs are acceptable for the intended use? What comparisons are needed to understand how much they affect the results?

3. Set the checks before release. Before making emissions available normally, finish the missing coverage, investigate the failed calculation check, and test the complete workflow through to a database export.

The accompanying notebook

The notebook starts by reading the existing verified database. It checks the data links and runs example calculations without rebuilding or writing a database. Separate options let the user repeat the build, create a new database, or preview the reviewed emissions work. A decryption key is requested only for a fresh build and is not saved. The unfinished mappings are excluded from the default setup.

Run instructions and a record of the supporting checks are in the same folder as this PDF. The full decision record is in docs/development/uvek-mapping-decisions.md.